6. This question is about the development of the Periodic Table.

In 1890, not all the elements had been discovered. A Russian scientist named Dmitri Mendeleev used the reactions of the known elements and their relative atomic masses to arrange them in the following table.

I	II	III	IV	V	VI	VII	VIII		
H 1.01									
Li 6.94	Be 9.01	B 10.8	C 12.0	N 14.0	O 16.0	F 19.0			
Na 23.0	Mg 24.3	Al 27.0	Si 28.1	P 31.0	S 32.1	Cl 35.5			
K 39.1	Ca 40.1		Ti 47.9	V 50.9	Cr 52.0	Mn 54.9	Fe 55.9	Co 58.9	Ni 58.7
Cu 63.5	Zn 65.4			As 74.9	Se 79.0	Br 79.9			
Rb 85.5	Sr 87.6	Y 88.9	Zr 91.2	Nb 92.9	Mo 95.9		Ru 101	Rh 103	Pd 106
Ag 108	Cd 112	In 115	Sn 119	Sb 122	Te 128	I 127			
Ce 133	Ba 137	La 139		Ta 181	W 184		Os 194	Ir 192	Pt 195
Au 197	Hg 201	Ti 204	Pb 207	Bi 209					
			Th 232		U 238				

Mendeleev's table

Based on the properties of the known elements, he predicted that new elements would be discovered and left gaps for these in his table. He even predicted what the properties of these undiscovered elements would be. He gave the name ekasilicon to one of them. The element that fits into this gap was eventually discovered and named germanium.

	Predicted properties of ekasilicon (Ek)	Actual properties of germanium (Ge)
Atomic mass	72	72.59
Density (g/cm^3)	5.5	5.35
Melting point	high	937.4 °C
Colour of metal	grey	grey-white
Formula of oxide	EkO_2	GeO_2
Formula of chloride	$EkCl_4$	$GeCl_4$

Comparison of the properties of ekasilicon and germanium

The modern day Periodic Table is based on Mendeleev's original table.

(a) (i) Put a tick (✓) in the correct column to show whether the following statements apply to Mendeleev's table only, today's table only or to both tables. [2]

	Mendeleev only	Today only	Both tables
the table is organised into groups			
copper and potassium are in the same group			
there are gaps in the table			
fluorine and chlorine are in the same group			

(ii) Tick (✓) the **two** statements that **best** describe why germanium was confirmed to be the element ekasilicon predicted by Mendeleev. [1]

germanium has exactly the same atomic mass as that predicted for ekasilicon ☐

germanium has a different colour to that predicted for ekasilicon ☐

germanium has a similar density to that predicted for ekasilicon ☐

germanium oxide has the same ratio of atoms as that predicted for ekasilicon oxide ☐

germanium oxide and germanium chloride have the same ratio of atoms ☐

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Examiner only

(b) (i) Calculate the percentage of oxygen present in the formula of germanium oxide, GeO_2. [2]

$A_r(Ge) = 73$ $A_r(O) = 16$

Percentage = ... %

(ii) When germanium oxide reacts with hydrochloric acid it produces germanium chloride and water.

Balance the following equation for the reaction that takes place. [1]

$$GeO_2 + \boxed{}\,HCl \longrightarrow GeCl_4 + \boxed{}\,H_2O$$

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